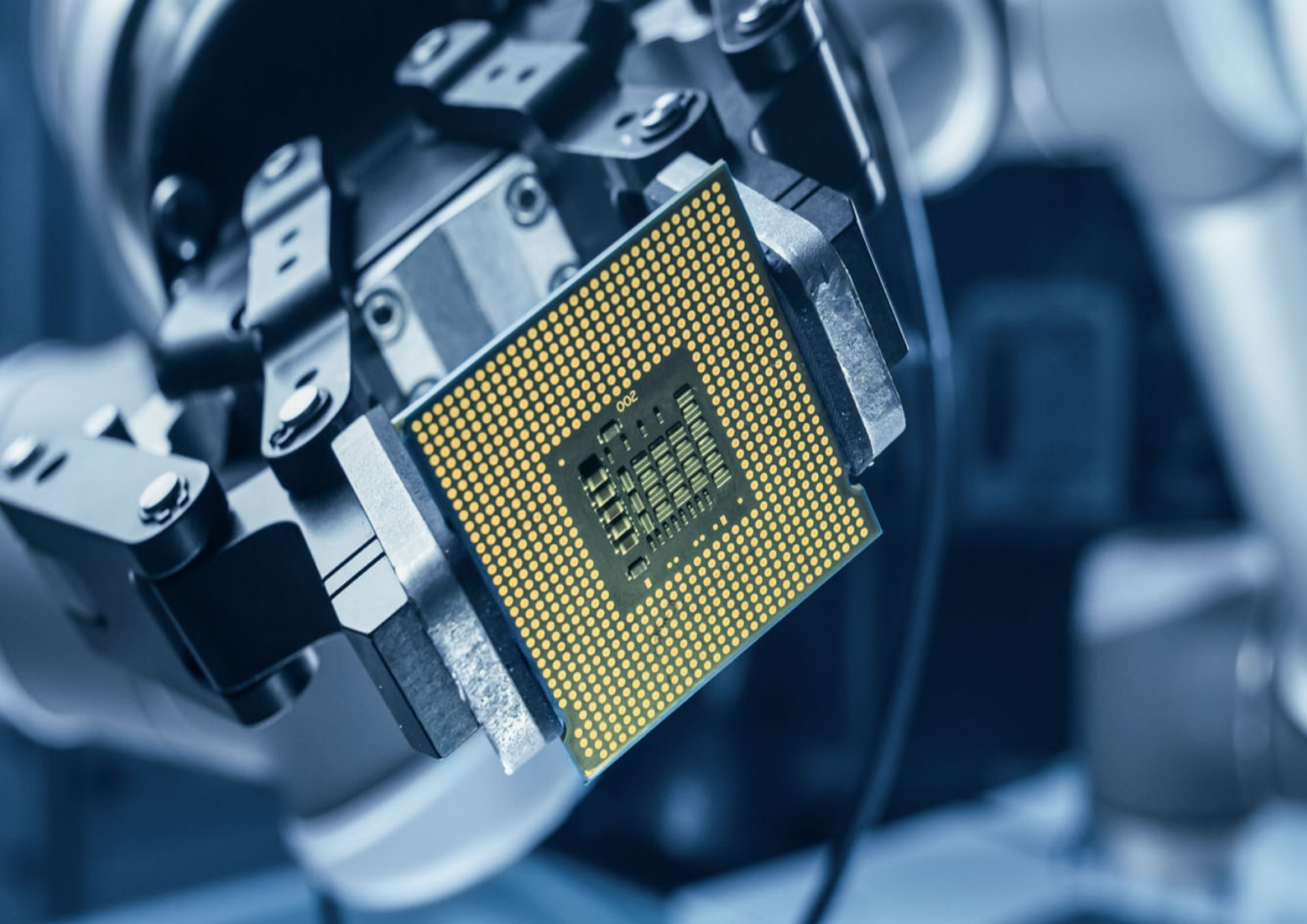




The SPICE Diode Model

Special Acknowledgment

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La corriente eléctrica

Sabemos que en situaciones electrostáticas (cargas eléctricas en reposo) no puede haber un campo eléctrico $\vec{E}$ en el interior de un material conductor, de otra manera las cargas móviles en el conductor se moverían y la situación ya no sería electrostática.

Una corriente (también llamada *corriente eléctrica*)

The SPICE Diode Model

The parameters for the SPICE model of the diode are given in Table 3.1. The equations that use these parameters are divided into four model groups which are responsible for simulation various diode characteristics. These groups consists of the large-signal dc model, the small signal ac model, temperature and area effects, and the noise model.

The Large-signal DC Model

The large-signal behavior of the SPICE diode is characterized by the relationship between the dc current and voltage at its terminals. The parameters used to model this behavior are IS, RS, N, BV y IBV. The parameters IS is the same as the reverse saturation current I_s for an ideal diode. The ohmic resistance RS is sued to model the resistance of the metal contacts and the neutral regions under high-level injection. The emission coefficient N is used to modify the slope of the current versus voltage (I-V) characteristic curve. Finally, the parameters BV and IBV model the reverse breakdown behavior. Figure 3.1 shows the equivalent circuits of the SPICE diode for large-signal dc analysis. This circuit contains an internal diode D_1 , a series resistance having a value of RS, and a shunt conductance GMIN. The SPICE program adds this conductance, which is transparent to the user, around every internal *pn* junction to aid convergence. The program default value for GMIN is 10^{-12} mhos but can be set to any other nonzero value with a .OPTIONS line in the circuit file.

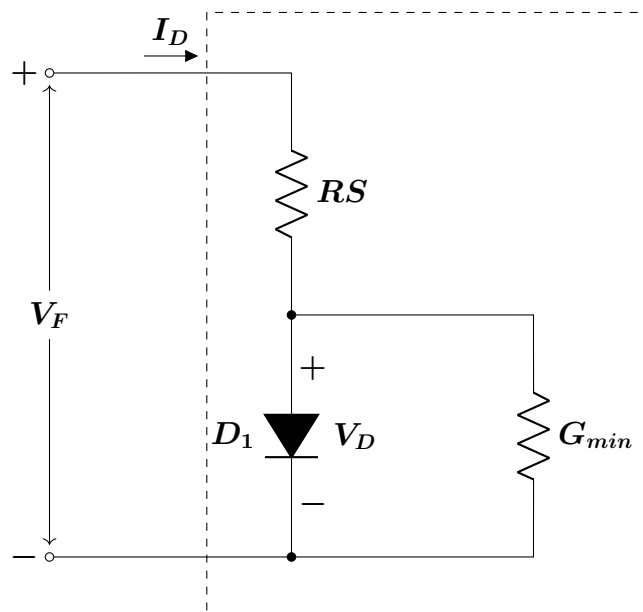


Figura 1: Amplificador multietapa

The dc model variables consists of the voltage across the external diode terminals V_F , the voltage across the interbal diode terminals V_D , and terminal current I_D . With this parameters and variables, the large-signal dc characteristics are modeled by the following equations.

$$V_F = RS \cdot I_D + V_D \quad (1)$$

y

$$I_D = f(V_D) \quad (2)$$

Where four regions of operation describe the functional relationship between the internal diode voltage and the diode current.

$$I_{C1} = \frac{\beta}{\beta + 1} I_{ref} \quad (3)$$

Donde

$$I_{ref} = \frac{V_{CC} - V_{BE}}{R_1} \quad (4)$$

Para $\beta \gg 1$

$$I_{ref} \cong I_{C1} \quad (5)$$

Ejemplo

Disele una fuente de corriente Widlar para generar una corriente constante $I_o = 10 \mu\text{A}$. Asuma que $V_{CC} = 10 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $\beta = 125$. Use $V_T = 25 \text{ mV}$.

Para la fuente de corriente Widlar primero debemos decidir un valor apropiado para I_{ref} . Si escogemos $I_{ref} = 1 \text{ mA}$, entonces R_1 está dado por

$$R_1 = \frac{10 \text{ V} - 0.7 \text{ V}}{1 \text{ mA}} = 9.3 \text{ k}\Omega \quad (6)$$

El valor de R_E está dado por la ecuación (7)

$$R_E = \frac{25 \text{ mV}}{\left(1 + \frac{1}{125}\right) 10 \mu\text{A}} \ln \left(\frac{1 \text{ mA}}{10 \mu\text{A}} \right) = 11.5 \text{ k}\Omega \quad (7)$$

Es claro ver que el circuito de Widlar permite la generación de constantes pequeñas usando resistores relativamente pequeños.